



ECONOMIC ANALYSIS AND MODELLING OF A HYBRID SOLAR- WIND DIESEL POWER GENERATION SYSTEM FOR RURAL ELECTRIFICATION

ADEBAYO DOSA, DAMILARE BABATUNDE AND KOLAWOLE ADISA

Email: dosbel@yahoo.com

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ABSTRACT

Energy plays a major task in the development of any nation's economy. This may not be unconnected to its inter-relationship with countries' national gross domestic product (GDP). In most developing countries, a reasonable percentage of national income initiates from the rural areas engaged in agricultural and related activities. Unfortunately, these communities depend predominantly on conventional sources of fuels which have detrimental environmental effects, as their source of supply of electricity. Consequently, there is little or no improvement in the standard of living in rural communities from their present low level without access to sustainable electricity. The aim of this study is to explore the prospect of meeting electricity demand from a solar wind-diesel hybrid system to a rural community isolated from the main utility grid in a coastal area of Nigeria. Based on the available renewable energy sources present at the site, a feasibility study has been carried out on the optimal hybrid renewable energy system (HRES) model to meet the electricity demand of a community of about 700 people. Hybrid Optimization Model for Electric Renewables (HOMER) is used to define the optimum configuration based on the available metrological resource. The outcome of the analysis is a list of feasible HRES configurations ranked according to their present cost (NPC). For this case study, the optimal configuration at the present interest rate of 12% in Nigeria is a combination of 20kW PV, 10 2.5kW, wind turbine, 50kW diesel generator, 17kW converter and 100 S4KS25P batteries. The NPC for this system configuration is N27,181,044 and LCOE is N71.676/kWh. The emission of CO₂ the principal pollutant can be reduced by 97.1% using HRES instead of standalone diesel generator. Furthermore, relationship of economic indices and the interest rate is also evaluated. In all, HRES though has a higher initial cost; it has a lower NPC, Levelized Cost of Energy (LCOE) and reduced greenhouse gas emission compared to diesel only generator.

KEYWORDS: PV modules, wind, battery, hybrid renewable energy systems, HOMER

INTRODUCTION

Decentralized and sustainable renewable energy systems have been promoted for electrification of remote areas where grid extension is not available or economically feasible (Emezirinwune et al., 2024b).

For providing electricity to somewhat populated remote areas such as Lagos, decentralized renewable energy systems providing electricity to households through hybrid systems could be considered as one of the options that is feasible and economically achievable (Javid et al., 2021).

Adebayo Dosa, Department of Industrial Engineering, Durban University of Technology, Durban, South Africa

Damilare Babatunde, Department of Meteorology and Climate Science, Federal University of Technology, Akure, Nigeria

Kolawole Adisa, Department of Electrical Electronic Engineering, University of Lagos, Nigeria

The hybrid systems also offer the possibility of integrating various locally available renewable resources to establish relatively larger generation capacity systems (Emezirinwune et al., 2024a). The rapid growth of industrialization and world population which results in the increase in energy demand, depletion of finite fossil fuel resources, and climate change, has made renewable energy resources increasingly attractive as an alternative to continued over-dependence on conventional energy sources (Babatunde et al., 2025). Studies have shown that renewable energy sources which include solar and wind energy are clean, inexhaustible and environmentally friendly, however, they are dependent on unpredictable factors such as weather and climatic conditions (Khan et al., 2021). The use of renewable energy helps in mitigation of climate change (Ataguba et al., 2024). Similarly, according to the Renewables Global Status Report for 2009, a stand-alone wind energy system does not produce usable energy for a considerable portion of time during the year due to relatively high cut in wind speed, which ranges from 2.5 to 4.5 m/s (Yolcan, 2023). These noted weaknesses of renewable energy resources, can partly be overcome by taking advantage of the complementary nature of the two energy sources, i.e. using a hybrid power system (Atawi et al., 2022). Hybrid power systems comprise at least two principal generators, usually a conventional generator powered by diesel and a renewable energy source such as wind, photovoltaic (PV), or 1 combination of PV and wind (Chrifi-Alaoui et al., 2023). Several studies on the feasibility, performance, and economic viability of hybrid power systems have been conducted and they found that hybrid power systems exhibit higher reliability and lower cost of generation than systems that employ only one source of energy (Ogunniran et al., 2025). Over the last decade, several studies on hybrid power systems have been conducted in different countries such as Denmark, Spain, Germany, United States, China, and India (Fragkos et al., 2021). Access to a reliable and stable supply of electricity is still a major challenge for both the urban and rural dwellers in Nigeria (Emezirinwune, 2022). Nigeria being endowed with abundant renewable energy resources such as solar and wind, hybrid power systems are expected to offer a unique possibility in electricity generation either as grid-connected or as stand-alone systems (Atawi et al., 2022).

In a country where huge amount of its population are rural dwellers and lack basic amenities such as electricity which can lead to poverty, unemployment, low standard of living and rural – urban migration it will be wise to find an alternative if not a lasting solution to this menace using a means that is suitable and economically viable to solve this global problem.

In Nigeria, availability of electricity and its cost has been a major concern to both the domestic and industrial consumers (Babatunde et al., 2017). At present, due to the fluctuations in the availability and maintenance of production sources that invariably lead to a shortfall in supply, access to grid electric power in Nigeria has generally been low (Ibrahim, 2023). This shortfall in electricity generation is responsible for the high cost of power supply and poses a serious challenge to both the manufacturing sector and the overall economic development of the country (Strielkowski et al., 2021). The reason for this high cost is the fact that on an average, 69% of the total energy required by the manufacturing companies is generated in house i.e. self-electricity generation using diesel generator (Shari et al., 2023). Even assuming that it is possible to generate the required peak demand of 12,390 MW, the national grid transmission capacity in Nigeria at the moment is less than 6000 MW and grid extension is capital intensive, especially in remote communities (Demirci et al., 2023). Similarly, standalone diesel generator sets are generally expensive to operate and maintain, especially at low load levels (Ramesh & Saini, 2024). In that respect, one of the options to overcome this pronounced energy shortage is by exploitation of hybrid sources of energy. Therefore, power generation that relies on isolated mini-grids and/or small (and large) scale off-grid distribution systems will help to improve the current level of electricity access in Nigeria.

MATERIALS AND METHOD

It is important to model a system mathematically before its optimal sizing. In the process of doing this, several techniques are developed. A concise description for the model for each system component is explained in the proceeding subsections. The system was simulated using the Homer software.

Mathematical Model for a Wind Generator System

There are several existing models for the estimation of wind turbine power (Bilendo et al.,

2022). Choosing a suitable model is very important for wind turbine power simulation. The Power output of a wind turbine (P_{wg}) can be modeled using Equation (1)

$$P_{wg} = \begin{cases} P_r \frac{V^2 - V_{cin}^2}{V_{rat}^2 - V_{cin}^2} & V_{cin} < V < V_{rat} \\ P_r & V_{rat} \leq V < V_{cou} \\ 0 & V \leq V_{cin} \text{ and } V \geq V_{cou} \end{cases} \quad (1)$$

where

P_r is the Rated power

V_{cin} is the cut-in wind speed

V_{rat} is the rated wind speed and

V_{cou} is the cut-off wind speed.

Power output of wind turbine generator at a specific site depends on wind speed at hub height and speed characteristics of the turbine. Wind speed at hub height can be calculated by using power-law equation (Liu et al., 2023). The equation used is

$$V = V_{ref} \left(\frac{H}{H_{ref}} \right)^\alpha \quad (2)$$

where

V is the wind speed (m/s) measured at the hub height H (m)

V_{ref} is the wind speed (m/s) measured at the reference height H_{ref} (m) and

α is the wind speed power law coefficient.

In order to use hybrid wind energy system effectively and economically, the chosen site should have good potential of wind energy. Moreover, technical feasibility and economic viability studies are to be carried out, in addition to capacity meeting of the demands (M. O. Babatunde et al., 20197). Capacity factor of wind turbine is also one of the deciding parameters to choose a particular type of wind turbine at the selected site, as an essential component of hybrid wind system (Omar et al., 2023. Optimal site planning for wind turbine installation also plays an

important role in wind power production (Emezirinwune et al., 2024a).

Mathematical Model for a Photovoltaic (PV) System

The photovoltaic module performance is highly affected by the weather, especially the intensity of the solar radiation and the PV module temperature. The input energy (I_T) to PV system is solar radiation and total solar radiation on an inclined surface is estimated using Equation (3).

$$I_T = I_b R_b + I_d R_d + (I_b + I_d) R_r \quad (3)$$

where

I_b is the direct normal solar radiation

I_d is the diffuse solar radiations,

R_d is the tilt factors for the diffuse part of the solar radiations

R_r is the tilt factors for the reflected part of the solar radiations

For estimating the PV generator output, the solar radiation available on the module surface, the ambient temperature and the manufacturers data for the PV module are used as model inputs. The calculation method is given below (Emezirinwune et al., 2024a).

The output power of the PV generator; P_g is given by the following equation

$$P_g = \eta_g \cdot A_g \cdot G_\beta \quad (4)$$

where

η_g is the PV generator efficiency

A_g is the total area of the PV generator (m²)

G_β is the solar radiation on tilted module plane (W/m²)

The tilt angle of the PV module is considered equal to the site latitude (Mukisa & Zamora, 2022). The PV generator efficiency is given by

$$\eta_g = \eta_r \eta_{pc} [1 - \beta(T_c - T_{cref})] \quad (5)$$

where

η_r is the reference module efficiency

η_{pc} is the power conditioning efficiency this efficiency is equal to 1 if a perfect maximum power point tracker is used,

β is the generator efficiency temperature coefficient

T_{cref} is the reference cell temperature (°C)

T_c is the cell temperature (°C) which can be estimated from the ambient temperature T_a (°C) and the solar radiation G_β follows

$$T_a + \left(\frac{NOCT - 20}{800} \right) G_\beta \quad (6)$$

where

$NOCT$ is the normal operating cell temperature (°C)

$T_{a,NOCT}$ is 20 °C

$G_{\beta,NOCT}$ is 800 W/m², for a wind speed of 1 m/s (Emezirinwune et al., 2024a).

Mathematical Model for a Diesel Generator

If the load requirements are not met by either renewable energy system or by batteries due to its state of charge, then the load requirements are met by operating the diesel generator in hybrid renewable energy systems HRES (Adebisi et al., 2023). The choice of diesel generator depends on type and nature of the load. To determine rated capacity of the engine generator to be installed,

following two cases should be considered. If the diesel generator is directly connected to load, then the rated capacity of the generator must be at least equal to the maximum load and if the diesel generator is used as a battery charger, then the current produced by the generator should not be greater than CAh/5 A, where CAh is the ampere hour capacity of the battery.

Overall efficiency of diesel generator is given by

$$\eta_{overall} = \eta_{brake\ thermal} \times \eta_{generator} \quad (7)$$

where

$\eta_{brake\ thermal}$ is the brake thermal efficiency of diesel engine

$\eta_{generator}$ is the generator efficiency

Mathematical Model for a Battery

During any hour, the excess power generated by the PV and wind generators can be utilized for charging the batteries whereas the stored energy can be discharged whenever there is a deficiency in power generation. When the power generated, by both the wind turbine and the PV array is insufficient and the storage is depleted, the load

will not be satisfied. Therefore, the difference between total energy generated and load demand energy, decides whether battery is in charging or discharging state. During the charging process, the available battery bank capacity at the time t can be calculated as follows (Emezirinwune et al., 2024a)

For the system configuration, the battery Capacity C_{bat} at time t is given by Equation 8

$$C_{bat}(t) = C_{bat}(t-1) + \left(P_{pv}(t)\eta_{cdd} + P_{acd}(t)\eta_{cw} - \frac{P_{load}(t)}{\eta_{inv}} \right) \eta_{cha} \Delta t \quad (8)$$

For the system configuration in which the wind turbine is taken as the primary source of energy, two cases can be considered.

When

$(P_{wg} \geq P_{load})$, the available battery capacity is given by

$$C_{bat}(t) = C_{bat}(t-1) + (P_{pv}(t)\eta_{cdd} + (P_{wg}(t) - P_{load}(t)) - \eta_{cad})\eta_{cha} \Delta t \quad (9)$$

When

$(P_{wg} < P_{load})$, the battery capacity is given by

$$C_{bat}(t) = C_{bat}(t-1) + \left(P_{pv}(t)\eta_{cdd} - \frac{P_{load}(t) - P_{wg}(t)}{\eta_{inv}} \right) \eta_{cha} \Delta t \quad (10)$$

Where

$C_{bat}(t)$ and $C_{bat}(t-1)$ is the available battery bank capacity in Watt hour (Wh) at the time t and $(t-1)$ respectively

$P_{load}(t)$ is the power consumed by the load at the time t

Δt is the simulation time step

η_{cdd} and η_{cad} is the DC/AC and AC/DC converter efficiencies respectively

The storage battery capacity is subject to the following constraints

$$C_{bat\ min} \leq C_{bat}(t) \leq C_{bat\ max} \quad (11)$$

where

$C_{bat\ max}$ is the maximum allowable storage capacities

$C_{bat\ min}$ is the minimum allowable storage capacities

$C_{bat\ min}$ is the determined by the maximum allowable depth of battery discharge (DOD)

$$C_{bat\ min} = DOD \cdot C_{bat} \quad (12)$$

The battery capacity is also dependent on temperature.

Electrical Circuit Model for combined Hybrid System

Figure 1 represents a basic electrical circuit model for the combined hybrid renewable energy system namely PV, wind, diesel, and battery. All the available renewable resources are connected to a

common node. The power from the sources can be utilized directly or through the battery (backup). For the basic model described, mathematical modelling can be performed based on either the power or current control. Power balancing is essential in the control system of the hybrid power systems configuration.

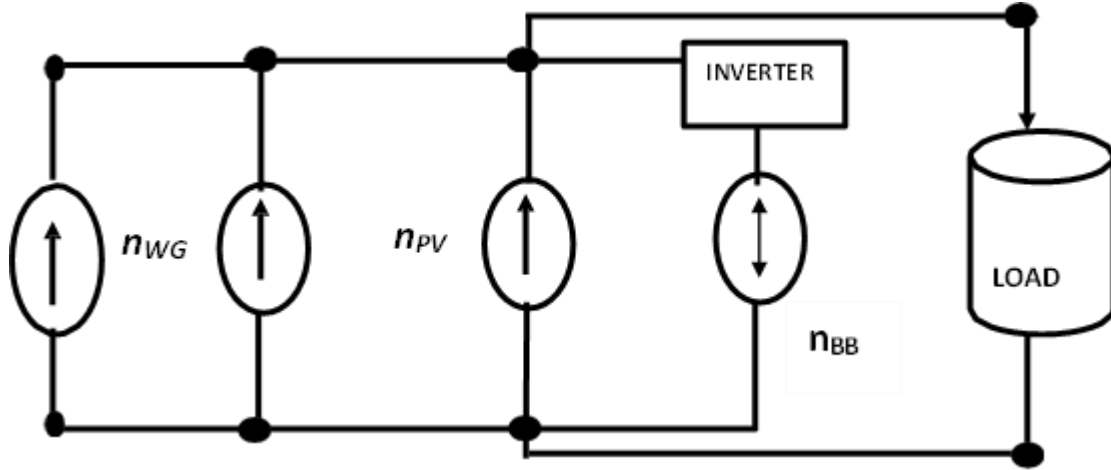


Figure 1: Electrical Circuit model for proposed hybrid configuration

From Figure 1, assuming the currents flowing out of η_{DG} , η_{WG} , η_{PV} and η_{BB} are i_{DG} , i_{WG} , i_{PV} and i_{BB} respectively, by Kirchhoff's Current law (KCL), the sum of the current i_{LD} entering the load is given as Equation 13.

$$i_{LD} = i_{DG} + i_{WG} + i_{PV} \pm i_{BB} \quad (13)$$

Since the sources are connected to a common node, the same voltage is expected across them. The sum of the power from the available sources is equivalent to that drawn by the load. Therefore, for a single plant of each type of available source the power balancing model is given as Equation 14:

$$P_{load}(t) = P_w(t) + P_{DG} + P_B + P_{PV} \quad (14)$$

where

$P_w(t)$ is the power of wind plant

P_{DG} is the power of diesel plant

P_B is the power supplied by battery

P_{PV} is the power supplied by PV panel

It is worth noting that the current can either flow in or out of the battery depending on if the battery is charging or discharging. When each source has more than one plant and the battery is supplying the bus, then the equation becomes:

$$P_{load}(t) = [P_w(t) + P_{DG}(t) + P_B(t) + P_{PV}(t)]X_K \quad (15)$$

When the battery is not serving as backup, then the expression reduces to equation (16)

$$P_{load}(t) = [P_w(t) + P_{DG}(t) + P_{PV}(t)]X_K \quad (16)$$

X_K is the number of each unit of the energy sources.

In Figure 1, the number of units of wind generators, number of PV unit and battery bank, and represented by n_{WG} , n_{PV} , and n_{BB} , respectively. The cumulative generated power feeds the loads through the AC bus. As earlier stated, the diesel generator is expected to operate between 80 and 100% of their kW rating. Surplus energy will be stored by the battery bank whenever the total generated power exceeds the load demand. The stored energy will be utilized when there is shortage of power to meet the load demand.

The state of charge of the battery bank at any time " t " is bounded above and below by a certain value SOC_{max} and SOC_{min} this is mathematically expressed as:

$$SOC_{min}(t) \leq SOC(t) \leq SOC_{max}(t) \quad (17)$$

The SOC_{max} is 1, SOC_{min} is a function of the maximum depth of discharge (DOD),

$$SOC_{min} = 1 - \frac{DOD}{100} \quad (18)$$

It is worth to note that equations (16) and (17) may operate under the following conditions:

- i. Combination of Wind generator and PV generator partly supplying the load and partly charging the battery banks
- ii. Hybrid wind/PV/diesel generator supplying the load.
- iii. Wind/PV generator partly supplying the load and dump load. While the battery is fully charged, the dump load dissipates the excess energy from the available sources.

Cost Modelling

Life cycle cost (LCC) is most times used to decide the economic feasibility of the system. The system with lowest LCC is always preferred. LCC is either calculated with or without accounting depreciation of the system by following the given equation

$$PV = \frac{\sum C_t}{(1+i)^t} + (C+m)_{wind} + (C+m)_{battery} + (C+m)_{diesel} + D \quad (19)$$

Where

$$PV_D = (C+m)_{PV}$$

PV and PV_D is the present value of the system without and with depreciation

t is the time of analysis

i is the interest rate per year

C_t is the cost in year t

m is the maintenance cost of the system

D is the present value of depreciation.

The cost per kWh of electricity is given by

$$\text{Cost, } \frac{\text{₦}}{\text{kWh}} = \frac{C_{actual}}{electricity} \quad (20)$$

where C_{actual} is the annualized cost and $electricity$ is the annual average generation [49].

Energy consumption Modelling

According to (Proedrou, 2021), the load profile can be estimated using equation (21)

$$HD(kWh) = \sum_t AHD \quad (21)$$

Where t is the hour under consideration (e.g. 12am-1am, 1am-2am). In this study t is taken as one hour interval.

But the appliance hourly demand (AHD) at any hour is given as:

$$AHD (kW) = DF \times \text{number of houses with appliance} \times \text{number of appliance per house} \times \text{wattage of equipment} \quad (22)$$

The demand factor is the ratio between the actual maximum load demand on the system to the total connected load. The demand factor varies from time to time and its value is always less than or equal to unity (Loth et al., 2022).

Hence;

$$\begin{aligned} \text{demand factor (DF)} &= \frac{\text{number of houses using appliance at that hour}}{\text{the total number of houses with such functional appliance}} \quad (23) \\ &= \frac{AHD (kW)}{\text{max. hourly load}} \end{aligned}$$

Designing and Modeling of Hybrid System with HOMER

Hybrid Optimization Model for Electric Renewables (HOMER) is an optimization software developed by the NREL (National Renewable Energy Laboratory) which performs economic analysis on proposed configuration (Sharma et al., 20213). HOMER, as micropower optimization model, simplifies the task of evaluating designs of both off-grid and grid-connected power systems for a variety of applications. When a power system is to be designed, certain decisions about the configuration of the system need to be considered. These include determining which components are appropriate to include in the design, the quantity of each component, and the optimal size for each.

The wide array of available technologies, variations in cost, and differences in energy resource availability can make this decision-making process complex. HOMER's optimization and sensitivity analysis algorithms help streamline this process by evaluating numerous potential system configurations to identify the most cost-effective and efficient solutions.

System configuration

As shown in Figure 2, the proposed hybrid system configuration is a combination of PV panels, wind turbines, diesel generators as well as storage batteries. The battery bank is needed for electricity storage. This stored energy can be accessed in times of shortage caused by, cloudy weather and night induced barriers.

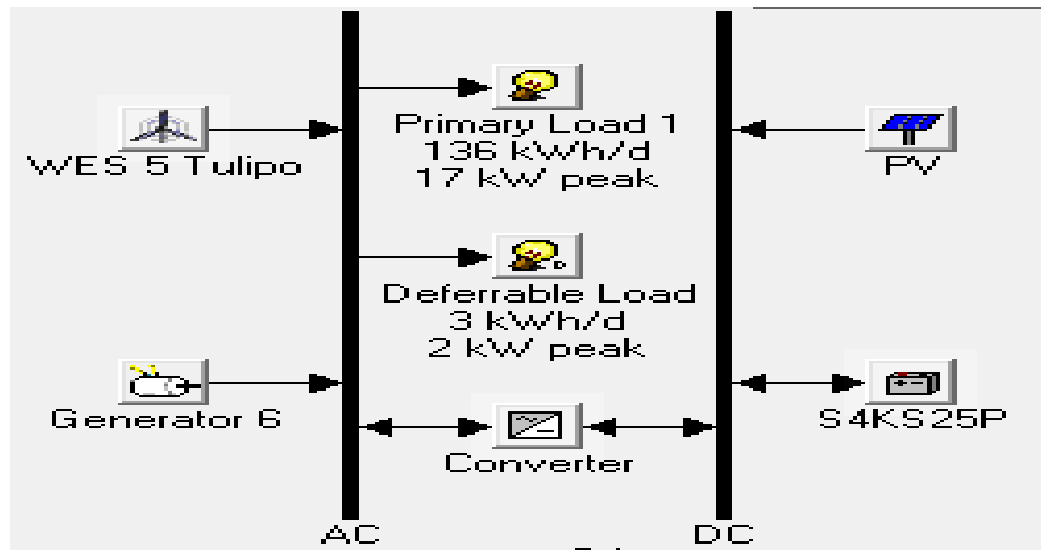


Figure 2: Hybrid system configuration

Input Parameters

Input parameters to HOMER includes, cost, solar and wind data, life time of project.

Resource Potential

The solar and wind data for the site under consideration was collected from two sources. These include the Nigerian Meteorological Agency

(NIMET) Oshodi and National Aeronautics and Space Administration (NASA) website. Figure 3 gives the monthly average daily solar radiation of the site under consideration for five years (2019-2024). The wind data obtained from NASA is shown in Figure 3.

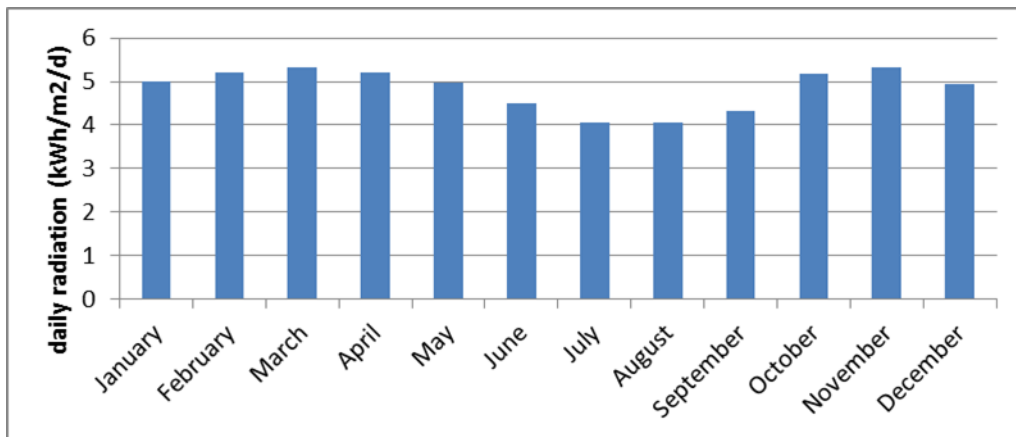


Figure 3: Monthly solar radiation

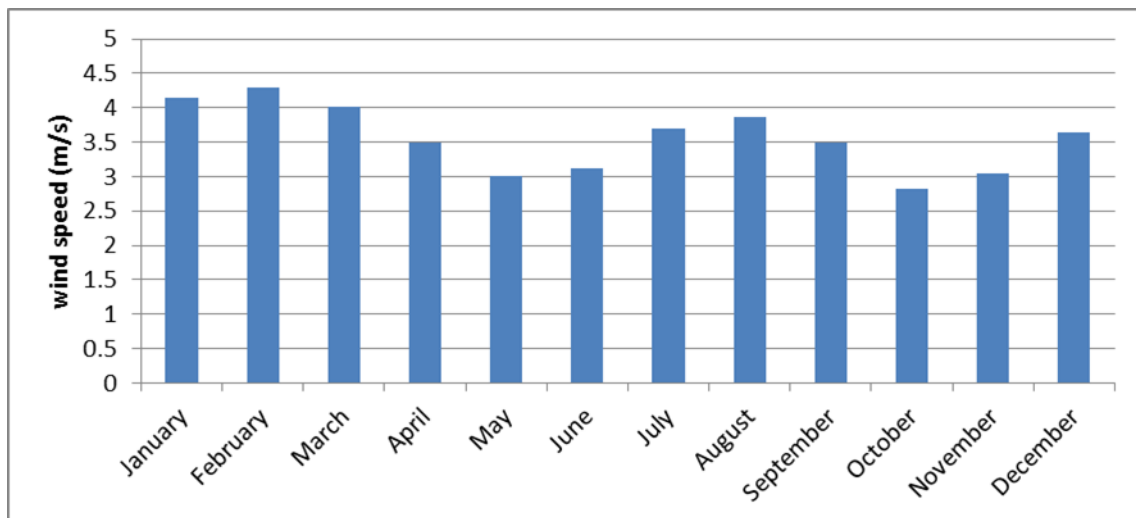


Figure 4: Wind resources

Simulation

The output power for the PV system is shown in Figure 4. The month of February has the highest power output while the month of July returned the least power output. According to Figure 5, the month of January returned about 13kW output of

power-the highest for the year, while the month of October recorded the least power output. The state of charge of the battery and the power output delivered by the diesel generator for the whole year is given in Figures 6 and 7 respectively

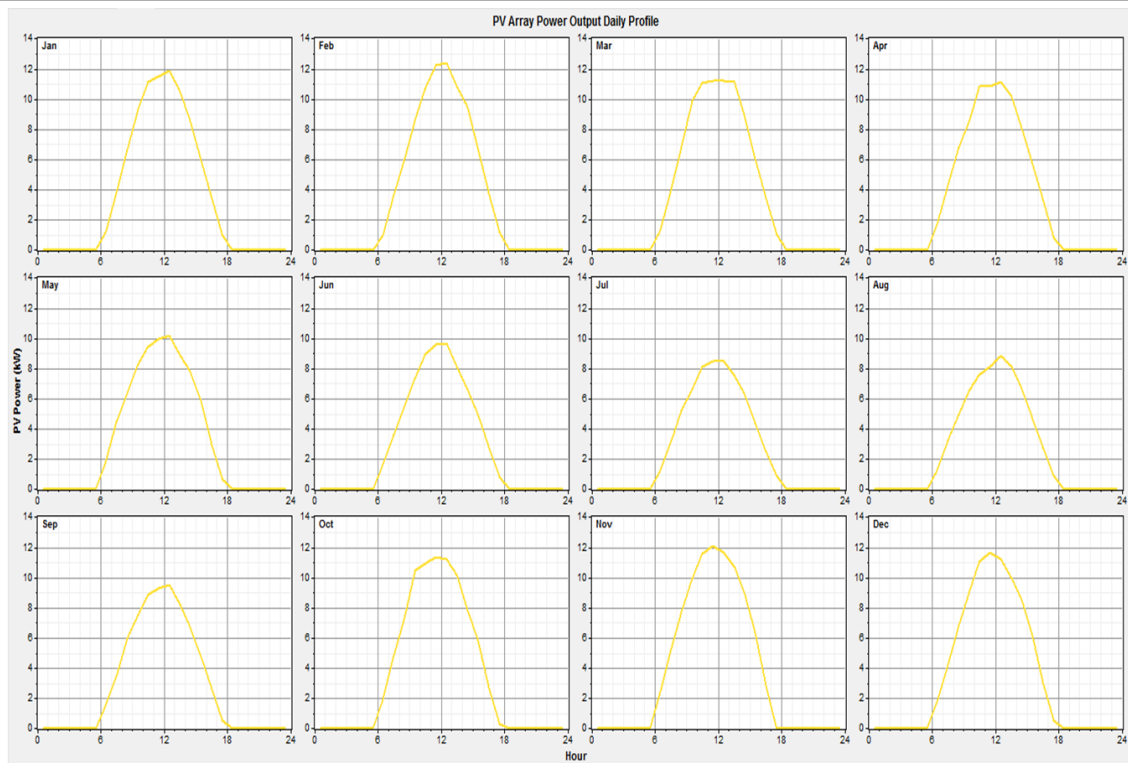


Figure 5: Power output form PV

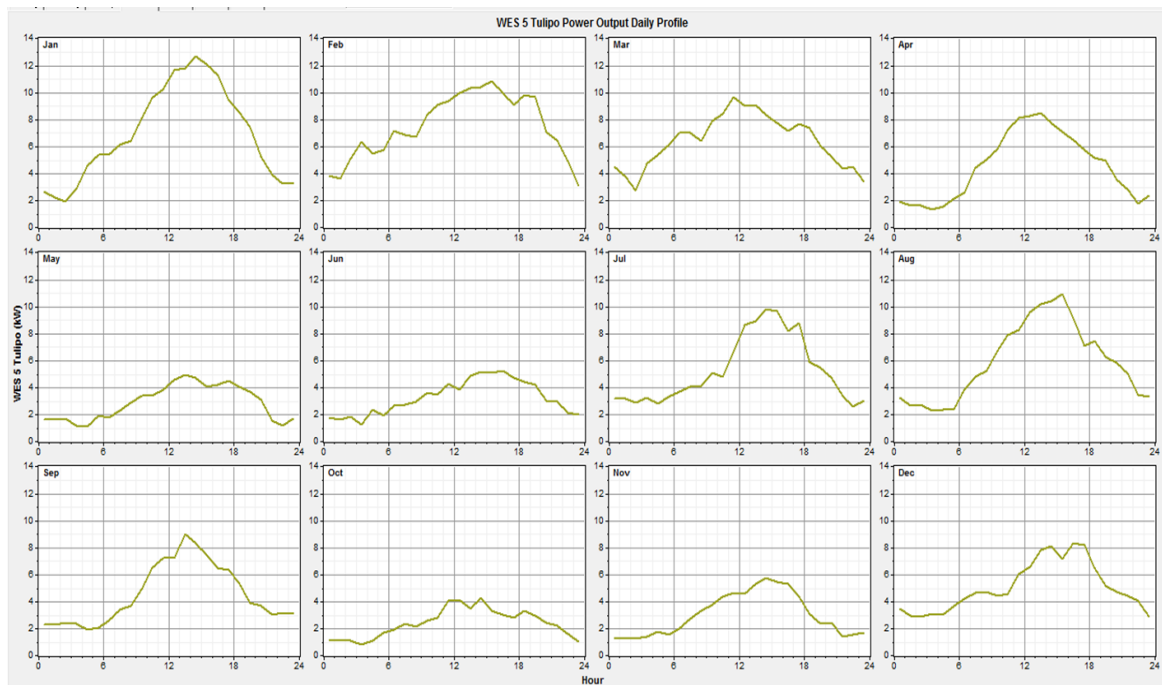


Figure 6: Power output form Wind Turbine

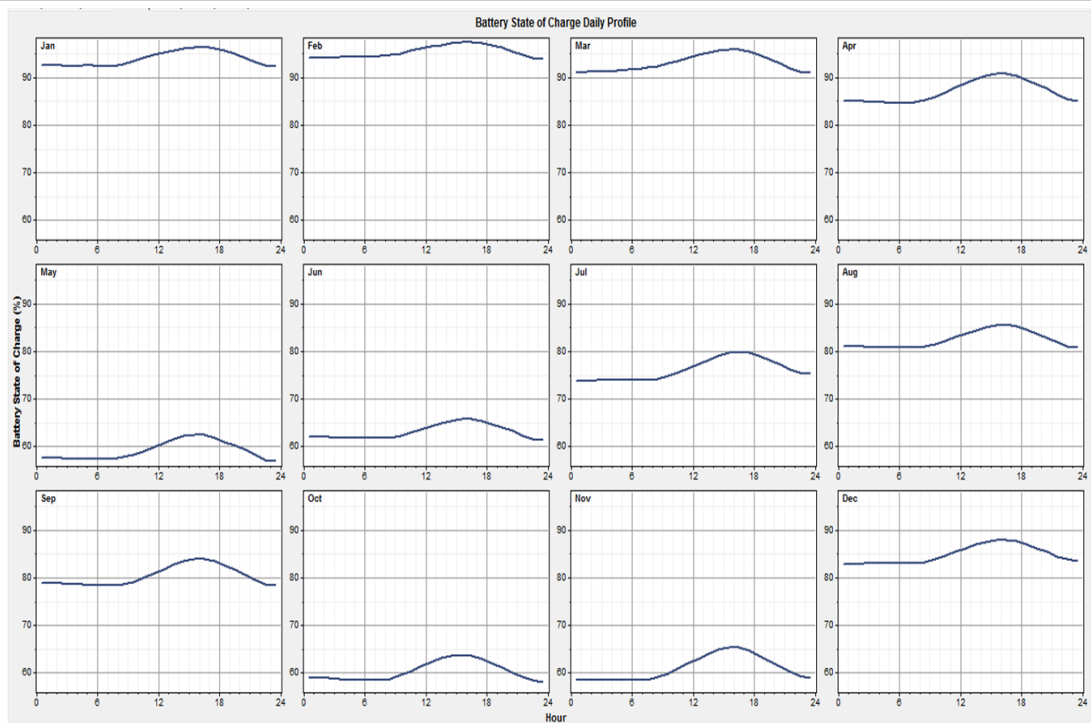


Figure 7: State of Charge of Battery

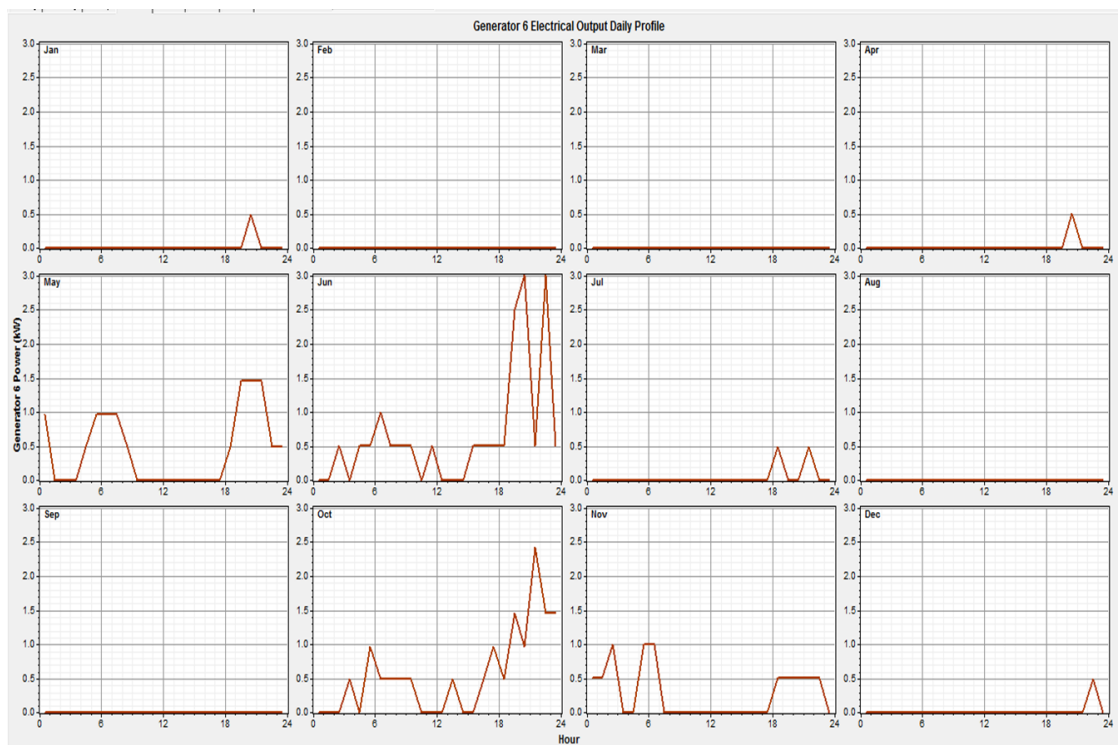


Figure 8: Power output of Diesel generator

RESULTS AND DISCUSSIONS

Using the available resources in the site under consideration, the hybrid energy system was simulated using HOMER. The load of the rural community was evaluated and employed as inputs to the system.

Load Profile

The load demand of Yovoyan village of Badagry varies on monthly basis. This is dependent on a number of factors such as holidays, Christmas, Easter, democracy day. These holidays keep people indoor thereby increasing electricity consumption during this period. Climatic conditions also contribute to the variation in energy consumption. The maximum peak load

occurred in January and December respectively. The monthly load profile of Yovoyan village is given in Figure 9 and Figure 10. Figure 9 shows the individual monthly load pattern; while Figure 10 shows that the peak load in the village is about 27kW with daily consumption estimated at 204 kWh/day. The primary load baseline as generated by HOMER is given in Table 1. The deferrable load pattern is represented by Figure 11. Deferrable load is electrical load that must be met within some time period, but the exact timing is not important. Loads are normally classified as deferrable because they have some storage associated with them. Water pumping is a common example - there is some flexibility as to when the pump actually operates, provided the water tank does not run dry. Other examples include ice making and battery charging.

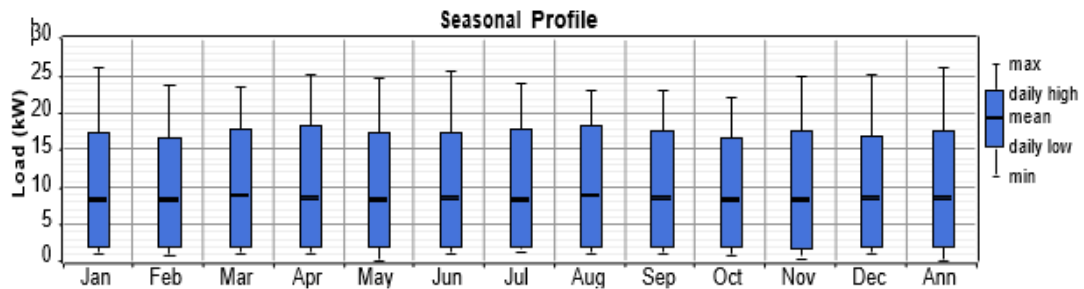


Figure 9: Individual monthly load pattern

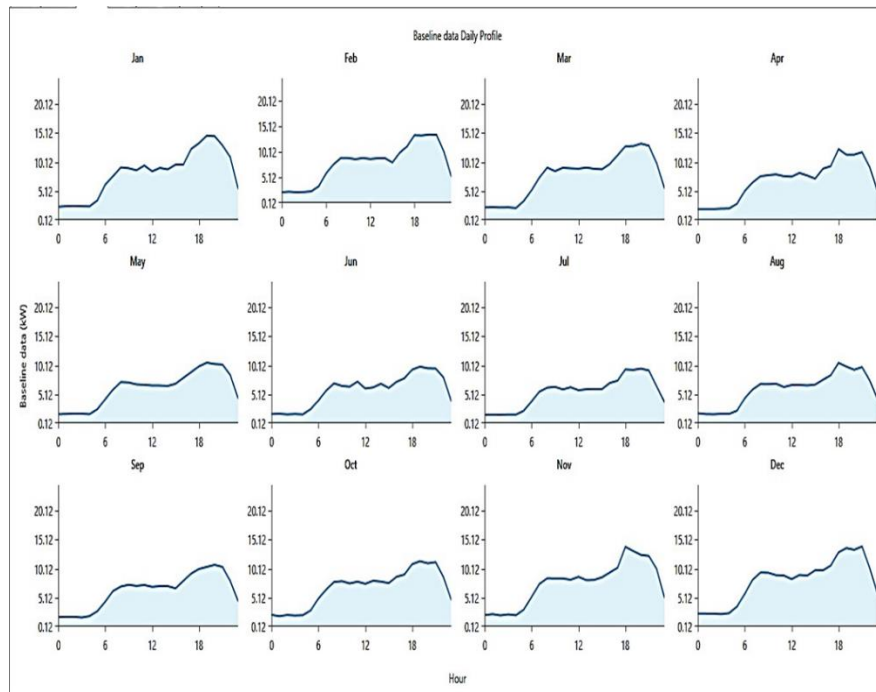


Figure 10: Monthly load profile for Yovoyan village

Table 1: Primary load baseline

	Baseline	Scaled
Average (kWh/d)	204	204
Average (kW)	8.5	8.5
Peak (kW)	26	26
Load factor	0.327	0.327

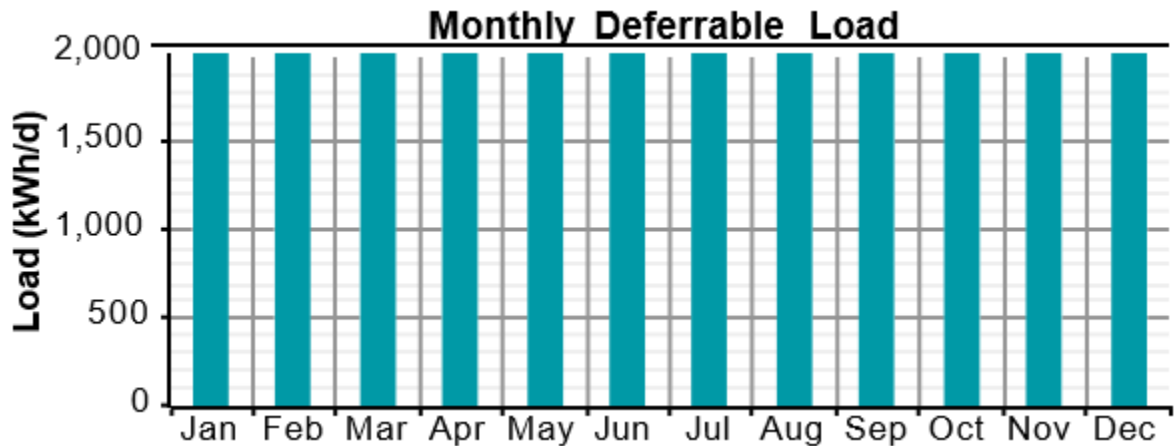


Figure 11: Monthly deferrable load profile for Yovoyan village

Optimization Results

The optimization results are shown in Table 2. A combination of PV/WIND/DG/BAT system configuration was returned as the best (serial number 1). According to the results, this system is made up of 20 kW PV module, a 50kW diesel generator, 17kW converter and 100 batteries with a dispatch strategy of load following. With this combination, renewable energy sources accounted for 97% of the total energy production. The diesel generator operates for 100 hours annually with the cost of energy as 71.28 N/kWh. This is chosen as the optimal based on the Net present cost of the system. The net present cost returned by this combination is the smallest of the combinations obtained. If the optimal choice is based on the total annual cost, the system configuration on serial number 2 will be the optimal. Also, based on the renewable energy fraction, total annual cost and net present cost, we will have to go for the system combination given by serial number 14. The COE for this configuration is N79.2/kWh.

Effect of Interest Rate on Economic Index

For the purpose of this research, the interest rate was varied from 0% to 12%. Zero (0%) was assumed if the government or utility provider can

secure an interest free loan from investors (e.g Islamic bank). The effect of the interest rate on the various economic indices such as initial cost, operating cost, total net present cost, and cost of energy is given in this section. The nominal cash flow through the project life time is given in Figure 12. The initial capital costs are usually paid at the beginning of the project. In the zeroth year, it is N-19,338,066. The replacement cost is dependent on the life time of the equipment. The battery has the shortest life time of 12 years. The break down summary of the net present cost is given in Figure 14.

Short term economic viability of the project

The advantages of using solar and wind resources to generate electricity include the avoidance of pollutant emissions, silent operation. The amount of annual solar energy reaching the Earth's surface is about 10,000 times more than annual global energy demand. Recently, in order to reach sustainable development, humankind needs to be steady on the path of low-carbon society. The feasibility study for the hybrid system is based on the findings of the wind and solar energy potentials at the particular locations. With the potentials determined, diesel generator is included in the model.

Based on Figures 20 - 23 it can be deduced that the advantage of the project is based on its long-term benefits. Therefore, the economic benefit comes from comparison to diesel power generator because we are considering a coastal region which is off-grid. In comparison to a diesel

generator, a hybrid system is not economically viable for the first five years of operation because of its initial cost. Though the running cost (fuel and maintenance cost) of the diesel generator is higher than that of a hybrid system, the hybrid systems' huge initial cost on the solar, wind and battery system while also incorporating the diesel generator makes it a very expensive alternative to the diesel generator.

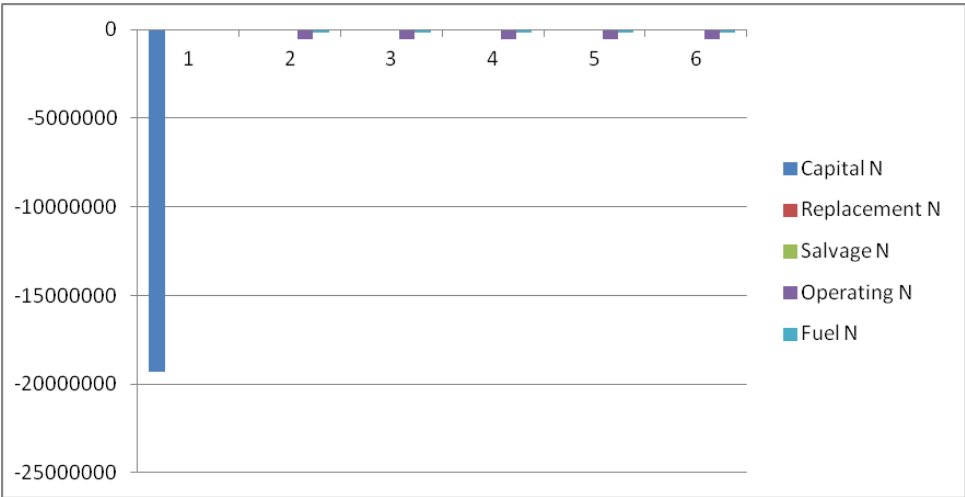


Figure 12: Nominal cash flows through the first 6 years of the project

Long term economic viability of the project

The generation of electrical energy through alternative sources such as wind and solar, has become more attractive and is widely used for substituting fossil fuels in the process of electrical power energy. Although PV systems are an expensive option of generating electricity when compared to other systems; this technology has been supported due to its potential benefits, which can be classified as customer-related benefits, electric utility related benefits and environmental benefits. The examples for the electric utility-related benefits are; reduced transmission and distribution costs and losses, peak shaving, and meeting peak demand. CO2 savings, NOX and SO2 savings can be listed as the environmental benefits of PV and wind systems. Despite the numerous alternatives, the choice is restricted by the varying net present cost of each set-up. Compared to the current global electricity tariff and the tariff in the country (< N20) the costs of the feasible set-ups obtained in this study are high, in the range of N71.676 to N79.2k per

kilowatt hour and this is slightly worrisome because of the notable difference in energy cost. These values have important implications for the Marginal Efficiency of Capital (MEC)—a critical economic index that reflects the expected return on additional investment relative to the cost of capital. In this case, the high cost per kilowatt-hour suggests a relatively low MEC, meaning the return on capital invested in these configurations may fall below the opportunity cost of capital or the discount rate used by investors. Consequently, private financiers and development agencies may view the project as economically unattractive unless policy measures—such as subsidies, concessional financing, or tax incentives—are introduced to raise the MEC to acceptable levels. Therefore, while the technical feasibility of these configurations has been demonstrated, the economic viability, as indicated by MEC, remains a significant hurdle. Investors are likely to demand higher expected returns or lower upfront capital costs to justify their participation in the deployment of these hybrid energy systems.

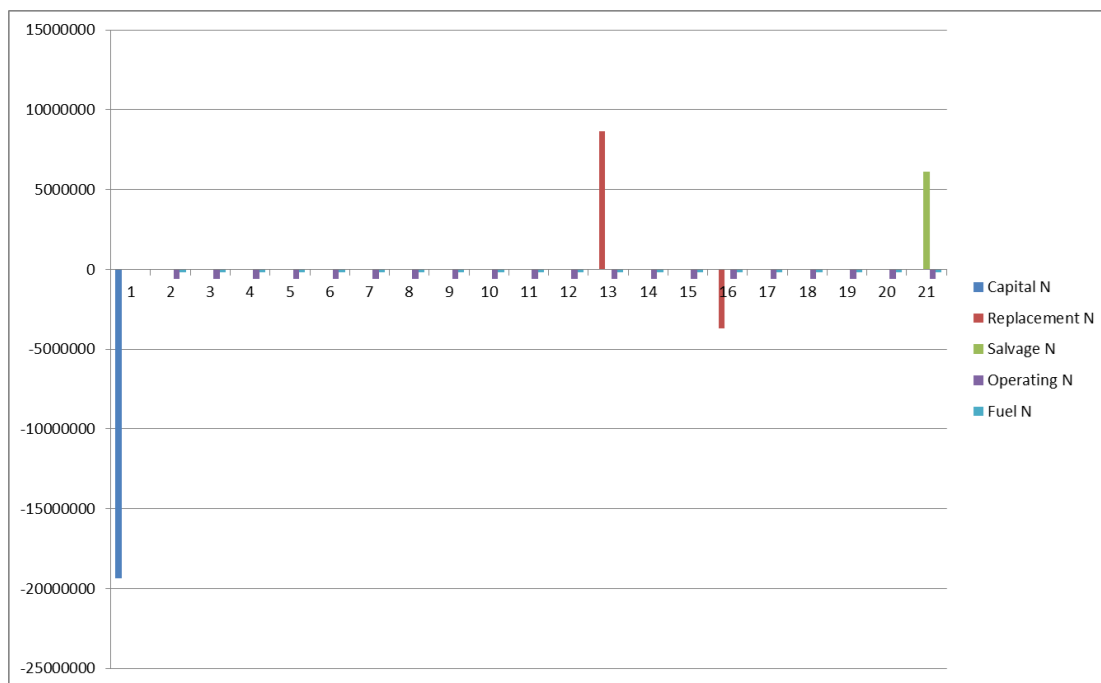


Figure 13: Nominal cash flows through the life time of the project

However, considering the shortage of power in the country (only 15 % coverage), its role in the improvement to the quality of life of the many people residing in the coastal region, the future situation regarding fossil fuel sources, and its contribution to the reduction of pollutant emissions into the environment, this cost should not be seen as a significant impairment. It should also be noted that free solar and wind energy will also be utilized, load will be satisfied in an optimal way; help is given to the mobilization of investments towards clean energy; and, most of all, the poor will benefit from the electricity provided. Although the net present cost is high, the running and maintenance costs are low as compared to the grid connection. With decreasing PV module prices, payback times on the PV based hybrid energy system investment are continuously decreasing.

Figure 13 illustrates the annualized distribution of major cost components—Capital, Replacement, Operating, Fuel, and Salvage—over a 21-year project lifecycle for a hybrid energy system. In Year 1, a significantly high capital expenditure

(over ₦20 million) is observed, representing the bulk of the initial investment required to deploy the system infrastructure. This large upfront cost sets the financial foundation for the project and influences the net present cost evaluation. Throughout the subsequent years (Years 2 to 12), operational and fuel costs remain relatively modest and consistent, suggesting that once operational, the system maintains stable recurrent expenditures. Notable spikes in replacement costs appear in Years 13 and 16, reflecting periodic equipment replacement, such as batteries or inverter units, which are typical for such systems. These spikes, though substantial, are expected and underscore the need for lifecycle planning and maintenance budgeting in renewable energy investments. In the final year (Year 21), a positive cash inflow is indicated in the form of a salvage value, amounting to approximately ₦6 million. This salvage recovery contributes to offsetting the long-term operational expenses and supports partial capital cost recovery.

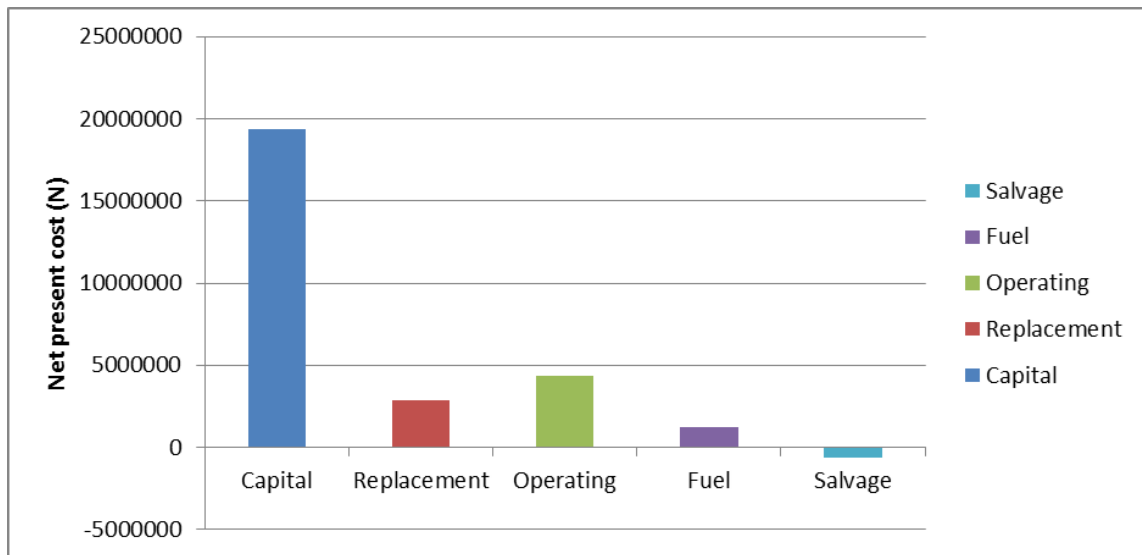


Figure 14: Cash flow summary by cost type

Table 2: Optimization Result

	PV	WES5	Generator	Surrette 4KS25P	Converter	Dispatch Strategy	Total Capital Cost					Total Fuel Cost	Operating Cost	COE	Ren. Fraction	Diesel		Label Hours
#	kW		kW		Kw		N	N	N/yr	N/yr	N/yr	N/yr	N/yr	N/kWh		L/yr	kg/yr	hr/yr
1	20	10	50	100	17	LF	19338066	27,181,044	303,336	577,764	168,894	168,894	1,049,994	72	0.97	775	2,041	100
2	15	15	50	100	17	LF	20080566	27,629,316	329,274	563,112	118,206	118,206	1,010,592	73	0.98	543	1,429	70
3	20	10	50	100	17	CC	19338066	27,838,800	303,732	586,476	247,896	247,896	1,138,104	73	0.95	1,139	2,998	114
4	20	15	50	100	17	LF	21194316	27,925,128	328,482	535,392	37,224	37,224	901,098	74	0.99	171	449	22
5	10	15	50	100	17	LF	18966816	28,253,016	330,858	623,700	288,684	288,684	1,243,242	75	0.95	1,325	3,490	171
6	15	15	50	100	17	CC	20080566	28,413,594	329,670	577,962	207,900	207,900	1,115,532	75	0.95	955	2,514	94
7	10	20	50	100	17	LF	20823066	28,476,756	355,608	562,914	106,326	106,326	1,024,650	75	0.98	488	1,286	63
8	10	15	50	100	17	CC	18966816	28,725,840	330,858	618,750	356,994	356,994	1,306,602	76	0.92	1,639	4,316	163
9	15	20	50	100	17	LF	21936816	28,876,716	354,816	538,758	35,442	35,442	929,016	76	0.99	163	429	21
10	20	15	50	100	17	CC	21194316	28,913,544	329,274	563,904	140,184	140,184	1,033,362	76	0.97	644	1,696	68
11	15	10	50	100	17	LF	18224316	29,123,226	306,306	685,278	467,478	467,478	1,459,062	77	0.92	2,147	5,653	277
12	15	20	50	100	17	CC	21936816	29,257,272	355,014	548,064	76,824	76,824	980,100	77	0.98	353	930	36
13	10	20	50	100	17	CC	20823066	29,317,464	356,004	581,328	199,782	199,782	1,137,312	77	0.96	918	2,416	93
14	20	20	50	100	17	LF	23050566	29,814,246	354,618	534,006	16,830	16,830	905,454	79	1	78	204	10

15	10	25	50	100	17	LF	22679316	29,949,678	381,348	546,480	45,540	45,540	973,368	79	0.99	209	551	27
16	15	10	50	100	17	CC	18224316	30,040,956	306,306	684,090	591,624	591,624	1,582,020	79	0.87	2,717	7,154	275
17	20	20	50	100	17	CC	23050566	30,237,174	355,014	545,094	61,974	61,974	962,082	80	0.99	285	750	28
18	10	25	50	100	17	CC	22679316	30,288,654	381,546	553,212	83,952	83,952	1,018,710	80	0.98	386	1,015	38
19	15	25	50	100	17	LF	23793066	30,852,558	381,150	540,342	23,562	23,562	945,054	81	1	109	286	14
20	15	25	50	100	17	CC	23793066	31,233,708	381,348	550,242	64,548	64,548	996,138	82	0.99	296	780	30
21	20	25	50	100	17	LF	24906816	31,824,936	380,952	536,778	8,514	8,514	926,244	84	1	39	102	5
22	20	25	50	100	17	CC	24906816	32,165,694	381,150	547,272	43,362	43,362	971,784	85	0.99	199	523	22
23	10	10	50	100	17	CC	17110566	33,022,044	309,474	807,048	1,013,760	1,013,760	2,130,282	87	0.78	4,654	12,256	477
24	10	10	50	100	17	LF	17110566	33,147,576	311,058	867,042	968,814	968,814	2,147,112	88	0.83	4,449	11,714	574
25	20	5	50	100	17	CC	17481816	35,121,834	284,526	857,142	1,220,076	1,220,076	2,361,546	93	0.74	5,602	14,751	561
26	20	5	50	100	17	LF	17481816	36,221,922	287,298	969,012	1,252,548	1,252,548	2,508,858	96	0.78	5,751	15,143	742
27	15	5	50	100	17	CC	16368066	39,936,996	292,050	1,026,432	1,836,648	1,836,648	3,155,328	105	0.6	8,433	22,207	838
28	15	5	50	100	17	LF	16368066	42,086,088	307,692	1,212,750	1,922,580	1,922,580	3,443,022	111	0.66	8,827	23,245	1,139
29	10	5	50	100	17	LF	15254316	49,268,538	331,650	1,502,820	2,719,332	2,719,332	4,553,802	130	0.52	12,485	32,878	1,611
30	20	10	50	300	17	LF	39138066	56,588,598	816,552	1,509,552	10,098	10,098	2,336,202	149	1	47	122	6
31	10	15	50	300	17	LF	38766816	57,123,000	843,480	1,536,282	77,616	77,616	2,457,576	151	0.99	357	939	46
32	20	10	50	300	17	CC	39138066	57,443,364	817,146	1,534,302	99,198	99,198	2,450,646	152	0.98	455	1,199	46
33	15	15	50	300	17	LF	39880566	57,557,610	842,886	1,513,512	10,098	10,098	2,366,496	152	1	47	122	6
34	15	15	50	300	17	CC	39880566	57,799,962	843,084	1,520,442	35,640	35,640	2,398,968	152	0.99	164	431	17
35	10	20	50	300	17	LF	40623066	58,491,774	869,220	1,516,284	6,732	6,732	2,392,236	154	1	31	82	4
36	10	20	50	300	17	CC	40623066	58,645,620	869,220	1,520,640	22,968	22,968	2,412,828	155	1	105	277	11

37	20	15	50	300	17	LF	40994316	58,686,012	842,886	1,515,492	10,098	10,098	2,368,476	155	1	47	122
38	20	15	50	300	17	CC	40994316	58,694,922	842,886	1,515,492	11,286	11,286	2,369,664	155	1	52	137
39	15	10	50	300	17	LF	38024316	58,756,500	819,522	1,625,184	330,858	330,858	2,775,564	155	0.94	1,519	4,000
40	10	15	50	300	17	CC	38766816	59,035,878	844,866	1,590,732	277,794	277,794	2,713,590	156	0.94	1,276	3,360
41	15	10	50	300	17	CC	38024316	59,320,800	819,522	1,620,828	410,850	410,850	2,851,200	157	0.91	1,886	4,967
42	15	20	50	300	17	LF	41736816	59,620,176	869,220	1,518,264	6,732	6,732	2,394,216	157	1	31	82
43	15	20	50	300	17	CC	41736816	59,620,968	869,220	1,518,264	6,930	6,930	2,394,216	157	1	31	83
44	10	25	50	300	17	LF	42479316	60,589,188	895,554	1,522,224	6,732	6,732	2,424,510	160	1	31	82
45	10	25	50	300	17	CC	42479316	60,605,622	895,554	1,522,224	8,910	8,910	2,426,688	160	1	41	108
46	20	20	50	300	17	LF	42850566	60,748,776	869,220	1,520,244	6,732	6,732	2,396,196	160	1	31	82
47	20	20	50	300	17	CC	42850566	60,749,370	869,220	1,520,244	6,930	6,930	2,396,196	160	1	31	83
48	15	25	50	300	17	LF	43593066	61,717,590	895,554	1,524,204	6,732	6,732	2,426,490	163	1	31	82
49	15	25	50	300	17	CC	43593066	61,734,222	895,554	1,524,204	8,910	8,910	2,428,668	163	1	41	108
50	10	10	50	300	17	CC	36910566	62,755,308	822,888	1,755,072	882,090	882,090	3,460,050	166	0.81	4,050	10,665
51	20	25	50	300	17	LF	44706816	62,846,190	895,554	1,526,184	6,732	6,732	2,428,470	166	1	31	82
52	20	25	50	300	17	CC	44706816	62,862,426	895,554	1,526,184	8,910	8,910	2,430,648	166	1	41	108
53	10	10	50	300	17	LF	36910566	62,988,948	824,472	1,814,472	852,390	852,390	3,491,334	166	0.85	3,914	10,306
54	20	5	50	300	17	CC	37281816	65,682,144	798,732	1,834,668	1,168,794	1,168,794	3,802,194	173	0.75	5,366	14,131
55	20	5	50	300	17	LF	37281816	66,150,216	800,910	1,919,412	1,144,440	1,144,440	3,864,960	175	0.8	5,255	13,837

56	15	5	50	300	17	CC	36168066	70,507,404	805,464	2,002,374	1,789,524	1,789,524	4,597,362	186	0.61	8,216	21,636
57	15	5	50	300	17	LF	36168066	72,015,372	819,126	2,163,744	1,816,254	1,816,254	4,799,124	190	0.68	8,339	21,959
58	10	5	50	300	17	LF	35054316	79,175,052	842,292	2,453,220	2,611,224	2,611,224	5,906,934	209	0.54	11,989	31,572

Table 3: Flow of Nominal Cost of Entire System

Component	Capital (N)	Replacement (N)	O&M (N)	Fuel (N)	Salvage (N)	Total (N)
PV	4,455,000	0	59202	0	0	4,514,202
Wind Turbine	3,712,500	633,006	88704	0	-239,382	4,194,828
Diesel Generator	990,000	0	2,334	1,260,864	-77814	2,635,182
Battery	9,900,000	2,220,966	3,697,452	0	-298,980	15,519,240
Converter	280,566	233	8316	0	-17424	317,592
System	19,338,066	2,900,106	4,315,806	1,260,864	-633,798	27,181,044

Levelized Cost of Energy and Total Net Present Cost

It can be seen from Figure 15 that the levelized cost of energy has a linear relationship with the interest rate. The levelized cost of energy (COE) as the average cost per kWh of useful electrical energy produced by the system. This implies that the best and optimal combination for minimum cost will be achieved if an interest free loan can be secured to execute the project. The minimum cost of energy for the best system returned by HOMER

is about N39.79 N/kWh while the maximum is around 67.04 N/kWh. The Total Net present cost decreases with increase in the interest rate. There are different methods of picking the optimal system configuration. These includes; NPC, COE and RE fraction. In order to strike a balance in our choice of system configuration, the interception point of both lines is taken as the optimal point. At this point, the TNPC is given as N44,550,000 at 49.5 N/kWh. The interest rate must therefore not exceed 6%.

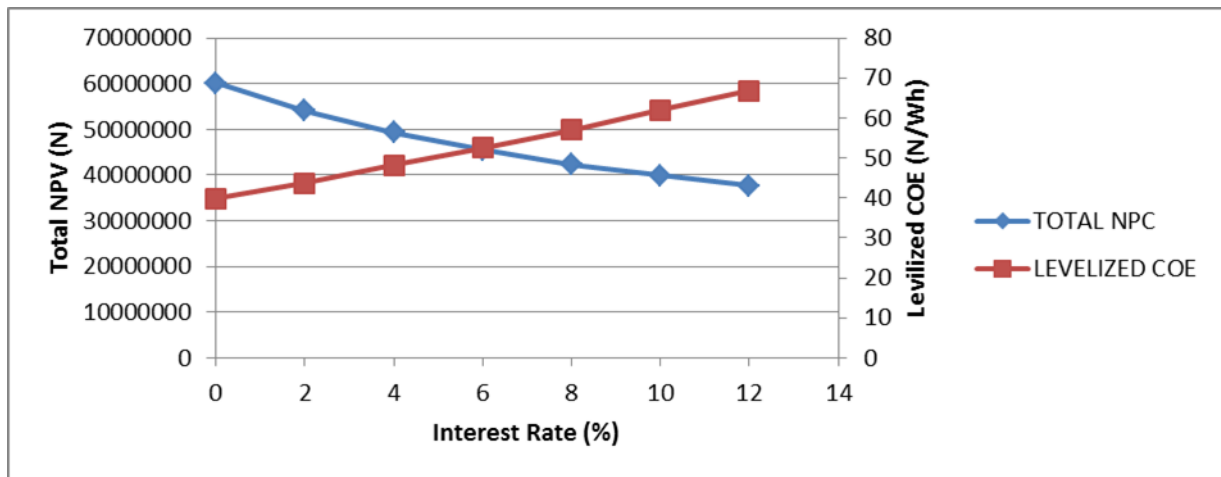


Figure 15: Relationship between NPV, LCOE and Interest rate

Operating Cost

The operating cost of the optimal system against the interest rate is shown in Figure 17. Initially the operating cost rises until the interest rate is 4%. The operating cost remained constant up to about

8% before dropping to N23,061,256 at about 10% interest rate. There was a rapid increase in the operation cost from 8% to 10% interest rate before finally dropping to N1,940,400/yr at 12%. The initial cost of the system was constant from 0% interest rate to 8% before dropping to N23,061,258 at 10%.

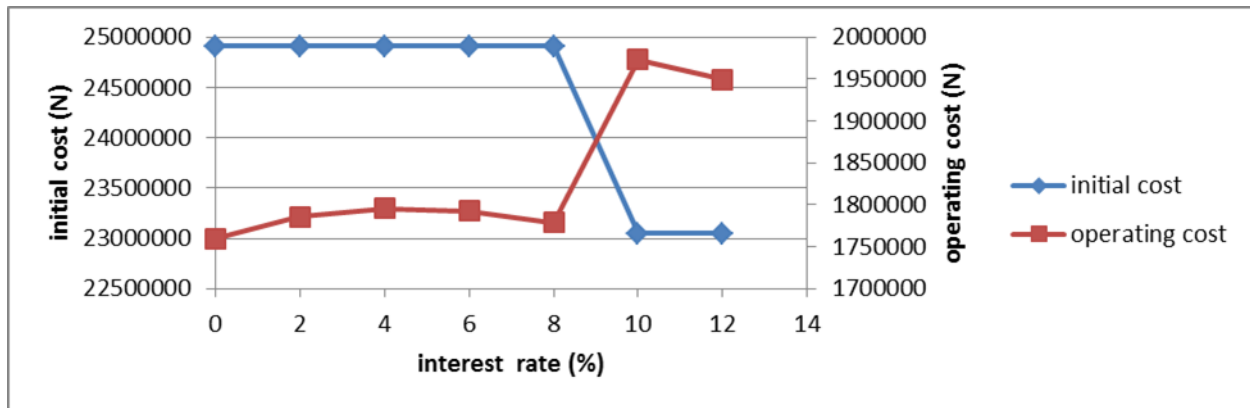


Figure 16: Relationship between Initial Cost, Operating Cost and Interest Rate

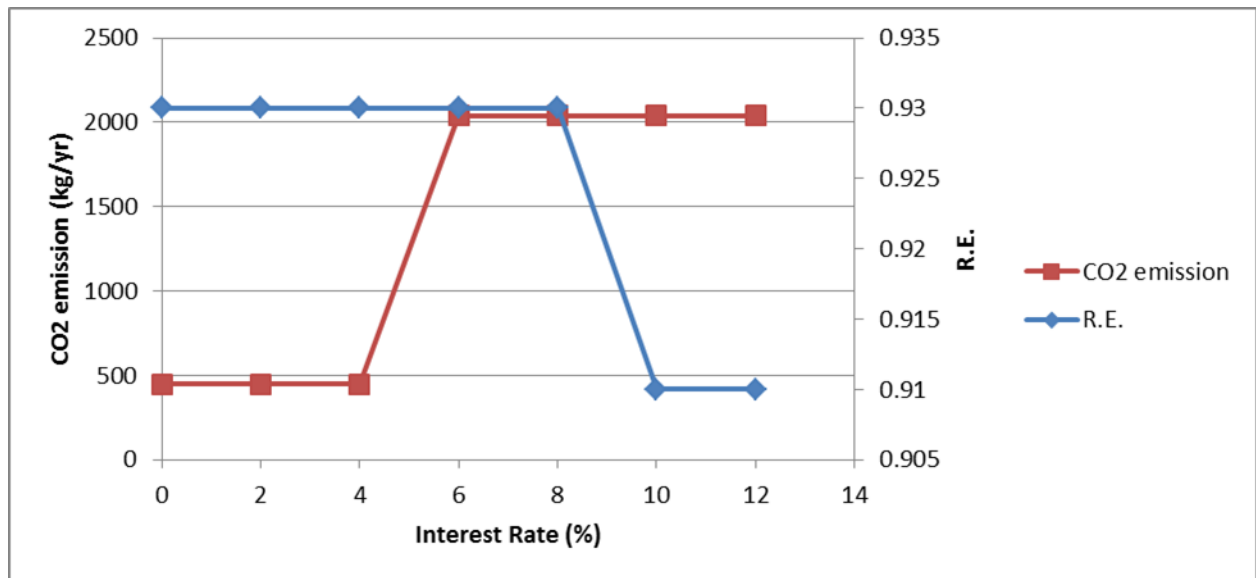


Figure 17: Relationship between CO2, RE and Interest rate

In Figure 17, at higher renewable energy fraction penetration, the CO2 emission was low due to the diesel generator operating at fewer hours. As the RE penetration decreases at 10 % interest rate, the emission increases. The relationship between

the fuel cost and the interest rate is shown in Figure 18, the fuel cost was constant between 0 to 4% interest rate but sharply increased to about N168,795/yr. for the rest of the incremental interest rate.

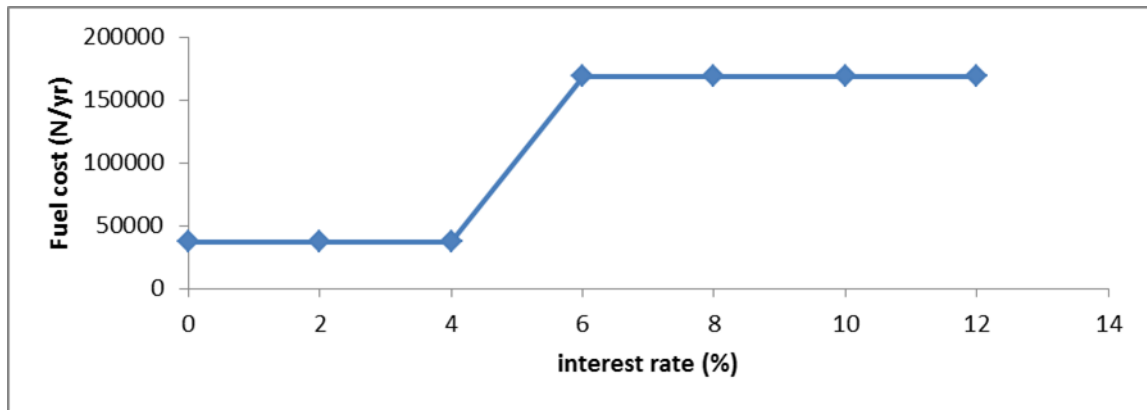


Figure 18: Relationship between Fuel cost and Interest rate

Electricity Production

The monthly percentage contribution to the total electricity produced by individual system component is given in Figure 21. For the system the PV supplied 40% of the annual electricity generated by the system, the wind turbine

supplied 58% while the diesel generator supplied the 2%. The months of May, June, October and November were the only months when the diesel generator was operational. These are the months when the wind resource is low.

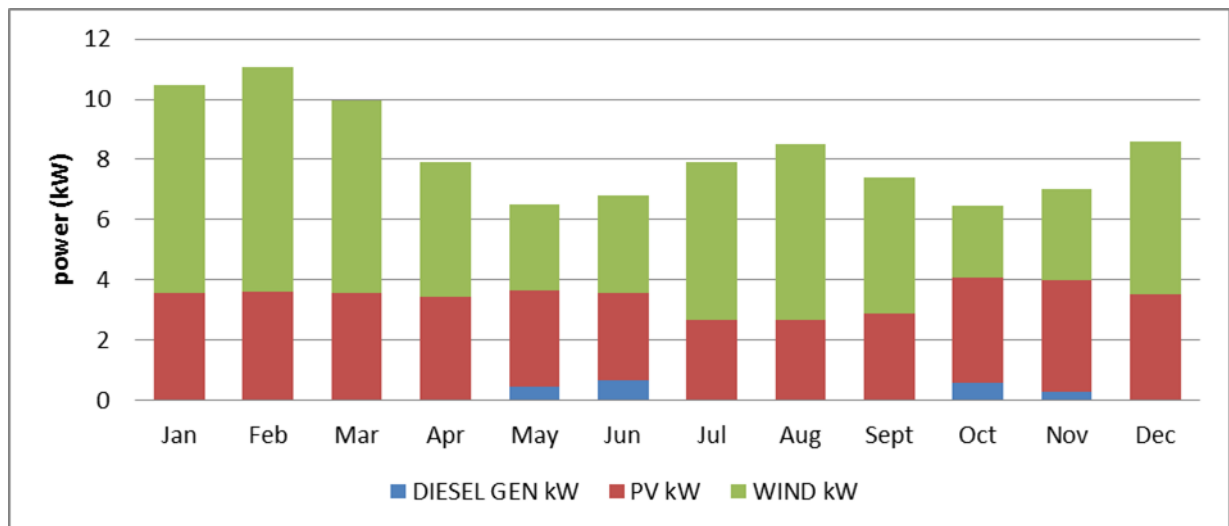


Figure 19: Electricity Production for HRES

Comparison of Standalone Diesel Generator and Hybrid Renewable Energy System

The comparisons between using standalone diesel engine and hybrid energy system in given

in Figures 19 – Figure 23. As shown in Figure 19, the net present cost of the standalone Diesel generator for supplying the community is higher than that of the HRES. This makes the HRES optimal.

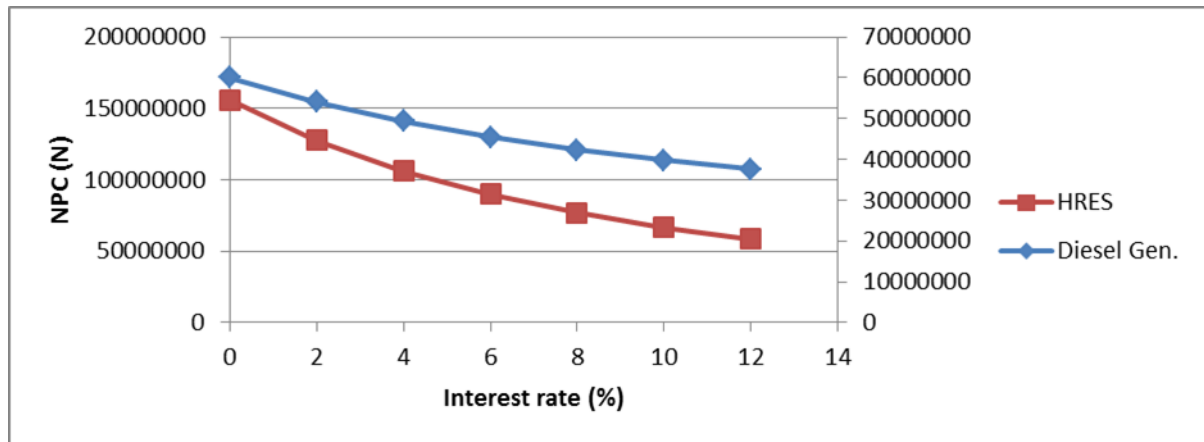


Figure 20: Comparison of NPC for Diesel Stand Alone and HRES

The initial cost comparison of the HRES and DG is shown in Figure 20. The DG is cheaper in terms of initial cost as compared to HRES. The initial cost of the DG is constant for all interest rates. In all, the DG has higher initial cost. The operating

cost of the HRES is lower as seen in Figure 21. The comparison of the LCOE as shown in Figure 22 shows that HRES has cheaper LCOE as compared to the standalone DG.

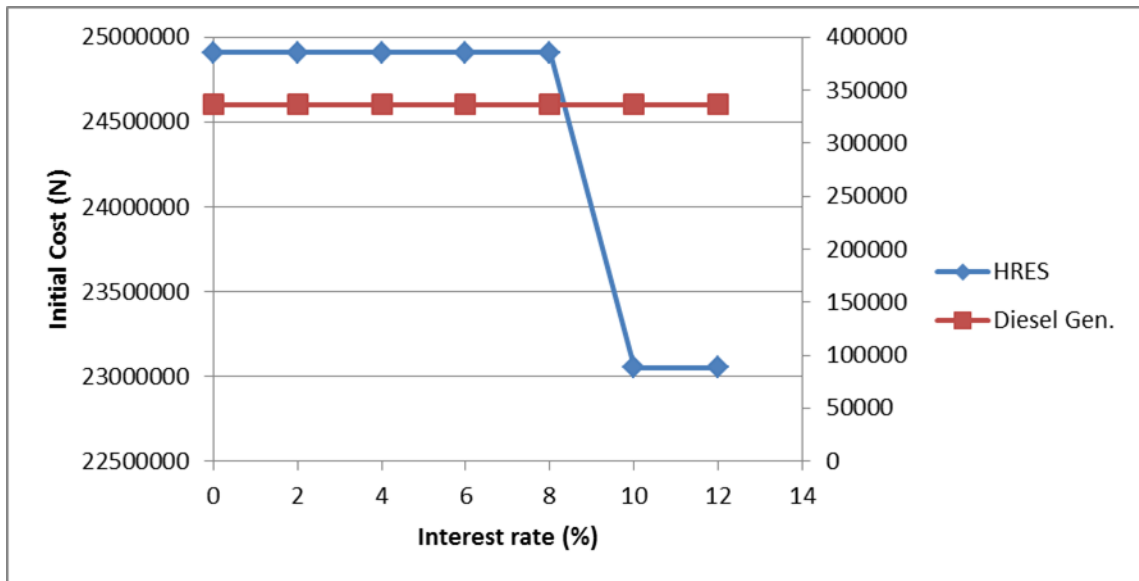


Figure 21: Comparison of Initial Cost for Diesel Stand Alone and HRES

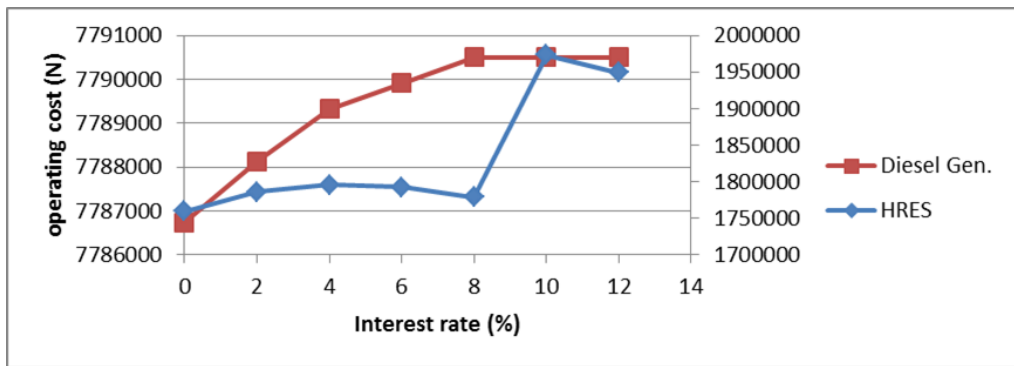


Figure 22: Comparison of Operating cost for Diesel Standalone and HRES

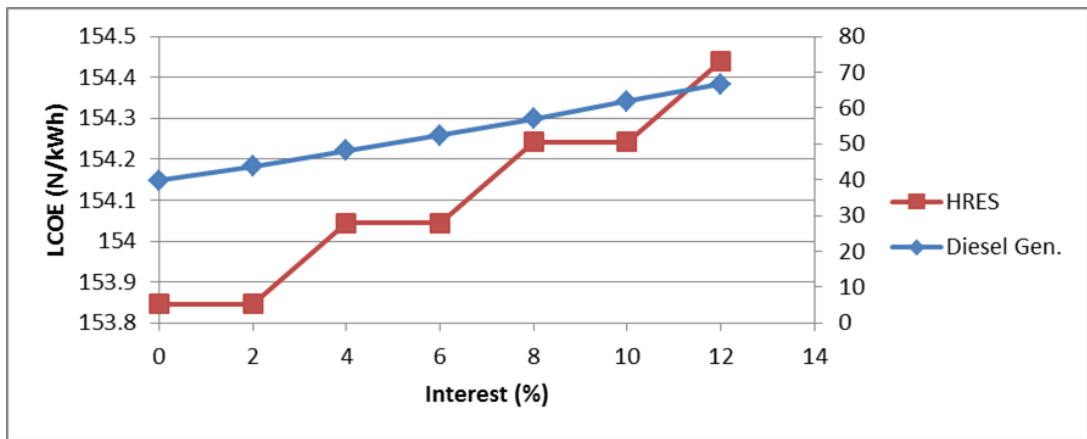


Figure 23: Comparison of LCOE for Diesel Stand Alone and HRES

Emission Analysis

Emission resulting from both optimal models for both DG only and HRES is given in Table 4. The most significant pollutant was CO₂ emission. Results show significant reduction in emission for the HRES model. CO₂ emission was reduced by

97.1% by using the HRES instead of the standalone diesel generator. This is due to 98% renewable energy penetration. Comparison of CO₂ released at each shortage capacity is given in Figure 24.

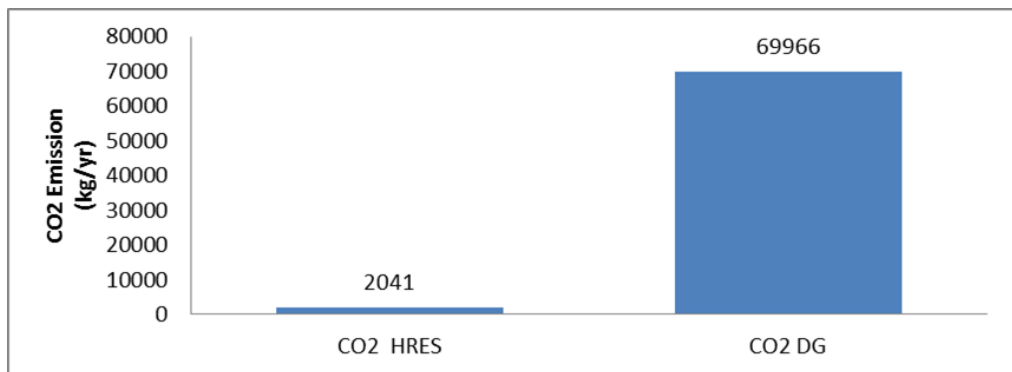


Figure 24: CO₂ emission comparison

Table 4: Emission Resulting from both Models

Pollutant	Emissions (kg/yr)	
	HRES	DG
Carbon dioxide	2,041	69,966
Carbon monoxide	5.04	173
Unburned hydrocarbons	0.558	19.1
Particulate matter	0.38	13
Sulfur dioxide	4.1	141
Nitrogen oxides	44.9	1,541

CONCLUSION

In this study, the optimal sizing of hybrid renewable energy system (HRES) for rural settlement located on a coastal area in Nigeria has been investigated. Based on the simulation results, the following can be concluded. Seasonal metrological data are very important to determine the applicable renewable energy potential of any location. This is either retrieved from local metrological agencies, measurement or from the NASA website. A 17-kW standalone diesel generator can be used to serve the rural load. For this model, the net present cost stands at N58,527,612 and LCOE as N154.44/kWh at 12% interest rate. Optimal configuration indicates the system with the most convenient component sizes to supply the primary load and deferrable load with the lowest total net present cost and cost of energy is the best system. For this case study, the optimal configuration at the present interest rate in Nigeria (12%) is PV-20kW, wind WES 10, Diesel generator-50kW, 17kW converter and 100 S4KS25P Batteries. Although the optimized hybrid renewable energy system (HRES) has a total net present cost of N27,181,044 and a levelized cost of electricity (LCOE) of N71.676/kWh, it remains cost-competitive when compared with the actual cost of self-generation or unreliable grid power in rural Nigeria. While Nigeria's official electricity tariff under the Multi-Year Tariff Order (MYTO) framework averages between N20–N65/kWh depending on customer class and region, rural and off-grid communities often experience severe power deficits, leading to reliance on diesel or petrol generators whose effective LCOE often exceeds N120–N150/kWh. In this context, the HRES offers a more stable and predictable

electricity cost, with significantly lower emissions and reduced fuel dependency. When compared to standalone diesel generation—which in this study has an LCOE of N154.44/kWh—the HRES demonstrates nearly 54% cost savings in energy generation, while simultaneously achieving a 97.1% reduction in CO₂ emissions. Thus, despite a higher initial capital outlay, the long-term economic and environmental benefits of HRES present a compelling case for rural electrification and decentralized energy planning in Nigeria.. It also shows that carbon emission can be reduced using HRES. In the meantime, this work is a further contribution to the potentials of HRES as a means of improving electricity supply without remarkable economic and environmental disintegration.

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